AIM :

To write a Python program to implement the Travelling Salesman Problem and find the minimum cost of the tour.

ALGORITHM :

1. Start the program.

2. Create a graph with cities and distances.

3. Generate all possible paths using permutations.

4. Calculate the total cost of each path.

5. Select the path with minimum cost.

6. Print the best path and minimum cost.

7. Stop the program.

PROGRAM :

prgm 9.py - C:/Users/nivet/AppData/Local/Programs/Python/Python314/prgm 9.py (3.14.6)

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```
from itertools import permutations

def tsp(graph, start):
    cities = list(graph.keys())
    cities.remove(start)

    min_cost = float('inf')
    best_path = []

    for path in permutations(cities):
        current_cost = 0
        current_city = start

        for city in path:
            current_cost += graph[current_city][city]
            current_city = city

        current_cost += graph[current_city][start]

        if current_cost < min_cost:
            min_cost = current_cost
            best_path = [start] + list(path) + [start]

    print("Best Path:", best_path)
    print("Minimum Cost:", min_cost)

# Graph representation
graph = {
    'A': {'B': 10, 'C': 15, 'D': 20},
    'B': {'A': 10, 'C': 35, 'D': 25},
    'C': {'A': 15, 'B': 35, 'D': 30},
    'D': {'A': 20, 'B': 25, 'C': 30}
}

tsp(graph, 'A')
```

OUTPUT :

```
========== RESTART: C:/Users/nivet
Best Path: ['A', 'B', 'D', 'C', 'A']
Minimum Cost: 80
>>>
```

RESULT :

Thus, the Python program to implement the Travelling Salesman Problem was executed successfully.